

NORTH SYDNEY BOYS HIGH SCHOOL

2011 YEAR 11 YEARLY EXAM

Mathematics

General Instructions

- Working time – 2.5 hours
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.

- Attempt all questions

Class Teacher:

☐ Mr Lin

Student Number: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	5	6	7	8	9	10	Total	Total
Mark	$\frac{10}{10}$	$\frac{10}{10}$	$\frac{14}{14}$	$\frac{12}{12}$	$\frac{10}{10}$	$\frac{10}{10}$	$\frac{11}{11}$	$\frac{11}{11}$	$\frac{13}{13}$	$\frac{8}{8}$	$\frac{109}{109}$	$\frac{100}{100}$

Question 1 (10 Marks)

Commence a NEW page.

Marks

- (a) Find the value of

$$\frac{13.81^2}{8.09 + \sqrt{4.62}}$$

Correct to (i) 2 decimal places

1

(ii) 1 significant figure

1

- (b) Fully factorise
- $8p^3 + 64$

2

- (c) Solve the following pair of simultaneous equations

3

$$2x - y - 7 = 0$$

$$3x + 2y - 14 = 0$$

- (d) Express the following surd with a rational denominator

3

$$\frac{2\sqrt{2}-3}{2\sqrt{2}+3}$$

Question 2 (10 Marks)

Commence a NEW page.

Marks

- (a) Simplify the following expression, giving the answer in index form:

$$2x^2y^{\frac{1}{3}} \times 3x^{-1}y^{\frac{5}{3}}$$

2

- (b) Given that
- $\log_2 3 = 1.58$
- and
- $\log_2 7 = 2.81$
- find the value correct to 2 decimal places of:

(i) $\log_2 49$

1

(ii) $\log_2 42$

2

- (c) Solve
- $3^x = 8$

2

- (d) Sketch the following graph clearly indicating the
- asymptote**
- and the
- intercepts**
- :

3

$$y = -\log_{10} x + 2$$

Question 3 (14 Marks)

Commence a NEW page.

Marks

$P(-3, 2)$, $Q(2, 1)$ and $R(-1, -2)$ are three points on the number plane.

- | | | |
|-----|---|---|
| (a) | Draw a neat sketch of the points on the number plane. | 1 |
| (b) | Show that the gradient of PQ is $-\frac{1}{5}$. | 2 |
| (c) | Show that the equation of PQ is $x + 5y - 7 = 0$. | 2 |
| (d) | Derive the equation of the line perpendicular to PQ passing through R . | 2 |
| (e) | Find the midpoint of PR . | 2 |
| (f) | Find the length of PQ . | 2 |
| (g) | Find the perpendicular distance of R from PQ . | 2 |
| (h) | Find the area of the triangle PQR | 1 |

Question 4 (12 Marks)

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Marks

- (a) Show whether the following functions are even, odd, or neither. Justify answers.

(i)	$f(x) = \sqrt{9 - x^2}$	2
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(ii)	$f(x) = x^3 - 4x + 5$	2
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(iii)	$f(x) = \frac{x^2}{x^2 + 4}$	2
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- (b) If $f(x) = 4x - 5$, Find

$\frac{f(a) + f(b)}{a - b}$	in its simplest form.	2
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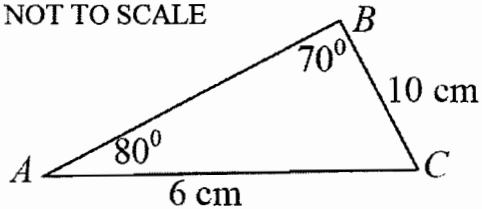
- | | | |
|---------|--|---|
| (c) (i) | Sketch the graph of $y = x^2 + x - 12$ | 2 |
| (ii) | Hence, or otherwise sketch the graph of $y = x^2 + x - 12 $ | 1 |
| (iii) | Hence, or otherwise state the domain and range of $y = x^2 + x - 12 $ | 1 |

Question 5 (10 Marks)

Commence a NEW page.

Marks

- (a) Write down the exact value of $\tan 300^\circ$. 2
- (b) Solve $2 \sin x + 1 = 0$ for $0^\circ \leq x \leq 360^\circ$. 2
- (c) NOT TO SCALE



- Find the area of the triangle ABC . 2
- (d) If $\sin \theta = -\frac{5}{13}$ and $\tan \theta < 0$, find $\cos \theta$ 2
- (e) Prove the identity 2

$$\tan \theta + \cot \theta = \frac{1}{\sin \theta \cos \theta}$$

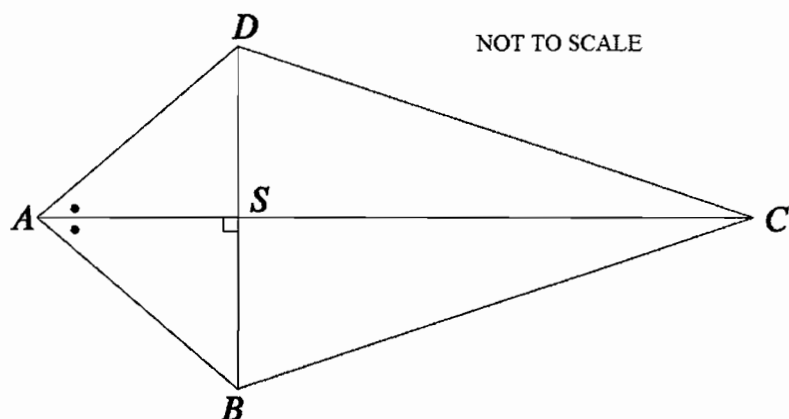
Question 6 (10 Marks)

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MarksDifferentiate the following with respect to x :

- (a) $y = 2x^4 - 8$ 1
- (b) $y = (2x - 1)^3$ 1
- (c) $y = x^2\sqrt{x}$ 2
- (d) $y = \frac{3}{2x^3}$ 1
- (e) $y = 5x^3(4x - 3)^5$ 2
- (f) $y = \frac{x^2 - 2}{x + 1}$ 3

(a)

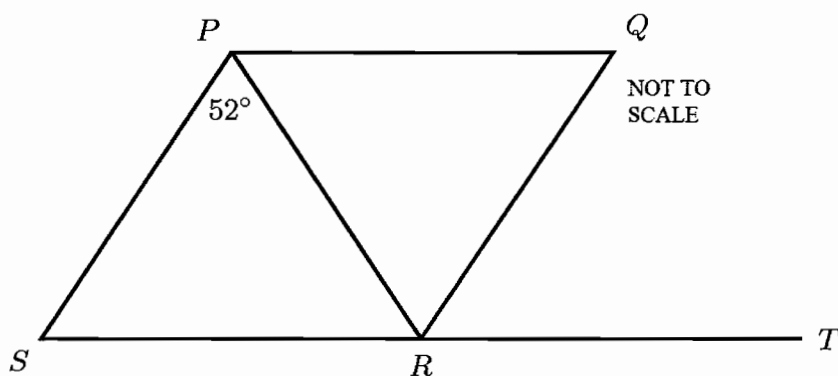


In the above diagram, $ABCD$ is a quadrilateral. The diagonals AC and DB intersect at right angles at point S . If $\angle DAS = \angle BAS$

(i) Prove that $\triangle ASB \equiv \triangle ASD$ 3

(ii) Hence prove that $DA = BA$ 1

(b) $PQRS$ is a rhombus where $\angle SPR = 52^\circ$ and SR is produced to T .



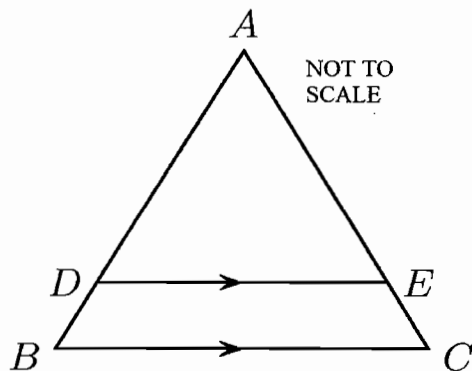
(i) Find the size of $\angle SPQ$. 1

(ii) What is the size of $\angle QRT$? Give reasons 2

Question 7 continued on next page...

...Question 7 continued

- (c) In the diagram below $DE \parallel BC$. $AB = 16\text{cm}$, $AE = 18\text{cm}$ and $EC = 6\text{cm}$.



- | | |
|--|---|
| (i) Prove that $\triangle ADE \parallel \triangle ABC$. | 2 |
| (ii) Find the length of DB | 2 |

Question 8 (11 Marks)

Commence a NEW page.

Marks

- (a) Find all values of x for which the tangent at x has positive gradient.

$y = x^3 - 3x$	3
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- | | |
|--|---|
| (b) Find the equation of the normal to $y = x^2 + 2x - 4$ at the point $(0, -4)$ | 3 |
|--|---|

- (c) From a point O , an observer can see a lighthouse L on a bearing of $285^\circ T$. An oil rig R can also be seen by the observer on a bearing of $215^\circ T$. The lighthouse is 12km from the oil rig and on a bearing of $012^\circ T$ from the oil rig.

- | | |
|---|---|
| (i) Draw a neat diagram showing the above information. | 1 |
| (ii) Find the distance, to the nearest kilometre, of the oil rig from the observer. | 3 |
| (iii) Find the bearing of the oil rig from the lighthouse. | 1 |

Question 9 (13 Marks)

Commence a NEW page.

Marks

If $f(x) = 6x^3 + 9x^2 - 3$,

(a) (i) Show that $6x^3 + 9x^2 - 3 = 3(x + 1)^2(2x - 1)$ 2

(ii) Hence find the x -intercepts 2

(b) Determine the y -intercept 1

(c) Find the stationary point(s) and determine their nature 4

(d) Find the point(s) of inflexion 2

(e) On the number plane sketch the curve 2

$$f(x) = 6x^3 + 9x^2 - 3$$

showing all of the above features

Question 10 (8 Marks)

Commence a NEW page.

Marks

A closed cylindrical can is made from 100π square centimetres of metal. If h is the height and r is the radius,

(i) show that $h = \frac{50}{r} - r$. 2

(ii) Hence, show that the volume can be expressed as $V = 50\pi r - \pi r^3$ 2

(iii) Find the maximum possible volume of the can and show why it is a maximum. 4

Answer to the nearest centimetres

END OF EXAMINATION

Question 1

a) i) 18.63

(ii) 20

b) $8(p^3 + 8) = 8(p+2)(p^2 - 2p + 4)$

c) $2x - y - 7 = 0$ — (1)

$3x + 2y - 14 = 0$ — (2)

$y = 2x - 7$ — (3)

sub (3) into (2)

$3x + 2(2x - 7) - 14 = 0$

$7x - 28 = 0$

$7x = 28$

$x = 4$

sub into (1)

$y = 1$

d) $\frac{2\sqrt{2}-3}{2\sqrt{2}+3} \times \frac{2\sqrt{2}-3}{2\sqrt{2}-3} = \frac{8 - 6\sqrt{2} - 6\sqrt{2} + 9}{8-9}$
 $= \frac{17 - 12\sqrt{2}}{-1}$
 $= 12\sqrt{2} - 17$

Question 2.

a) $2x^2 y^{\frac{1}{3}} \times 3x^{-1} y^{\frac{5}{3}} = 6x y^2$

b) i) $\log_2 3 = 1.58$, $\log_2 7 = 2.81$

i) $\log_2 7^2$

$= 2\log_2 7$

$= 2 \times 2.81$

$= 5.62$

ii) $\log_2 42 = \log_2 (6 \times 7)$

$= \log_2 3 + \log_2 2 + \log_2 7$

$= 1.58 + 1 + 2.81$

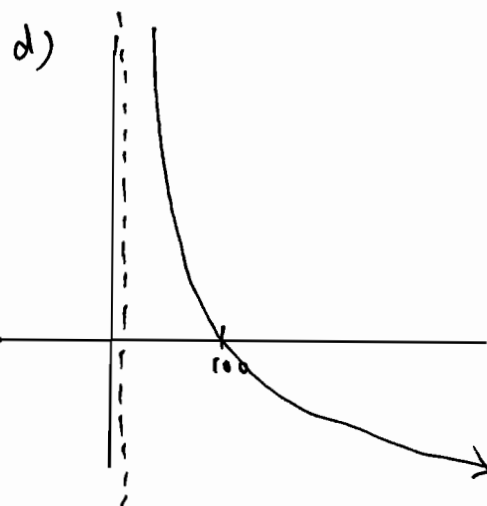
$= 5.39$

c) $3^x = 8$

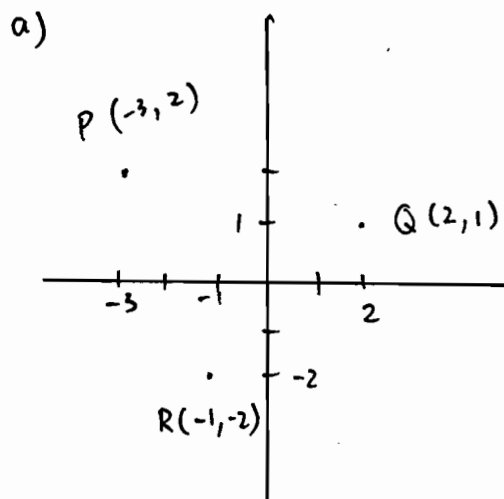
$\log_3 8 = x$

$x = \frac{\log 8}{\log 3}$

$= 1.89$ to 2dp.



Question 3



$$b) m_{PQ} = \frac{2-1}{-3-2} = \frac{1}{-5}$$

$$= -\frac{1}{5}$$

$$c) PQ \Rightarrow$$

$$y-2 = -\frac{1}{5}(x+3)$$

$$5y - 10 = -x - 3$$

$$x + 5y - 7 = 0$$

$$d) \text{ gradient of normal}$$

$$= -\frac{1}{-\frac{1}{5}}$$

$$= 5$$

$$\therefore y + 2 = 5(x + 1)$$

$$5x - y + 3 = 0$$

$$e). \left(\frac{-3-1}{2}, \frac{2-2}{2} \right)$$

$$M(-2, 0)$$

$$f) \sqrt{(2-1)^2 + (-3-2)^2}$$

$$= \sqrt{1^2 + 25}$$

$$= \sqrt{26}$$

$$g) \frac{|(-1 \times 1) + (5 \times -2) - 7|}{\sqrt{26}}$$

$$= \frac{|-18|}{\sqrt{26}}$$

$$= \frac{18}{\sqrt{26}}$$

$$h) A = \frac{1}{2}bh$$

$$= \frac{1}{2} \times \sqrt{26} \times \frac{18}{\sqrt{26}}$$

$$= 9$$

Question 4

$$a) i) f(x) = \sqrt{9-x^2}$$

$$f(-x) = \sqrt{9-(-x)^2}$$

$$= \sqrt{9-x^2}$$

$$f(x) = f(-x)$$

$\therefore$ even

$$(ii) f(x) = x^3 - 4x + 5$$

$$f(-x) = (-x)^3 - 4(-x) + 5$$

$$= -x^3 + 4x + 5$$

$\therefore$ not even

$$-f(x) = -x^3 + 4x - 5$$

$$\neq f(x) \therefore \text{not odd}$$

$\therefore$ neither

$$(iii) f(x) = \frac{x^2}{x^2+4}$$

$$f(-x) = \frac{(-x)^2}{(-x)^2+4}$$

$$= \frac{x^2}{x^2+4}$$

$$= \frac{x^2}{x^2+4}$$

$$= f(x)$$

$\therefore$ even

$$4(b) \quad f(a) = 4a - 5$$

$$f(b) = 4b - 5$$

$$\therefore \frac{f(a) + f(b)}{a - b}$$

$$\Rightarrow \frac{4a - 5 + 4b - 5}{a - b}$$

$$= \frac{4a + 4b - 10}{a - b}$$

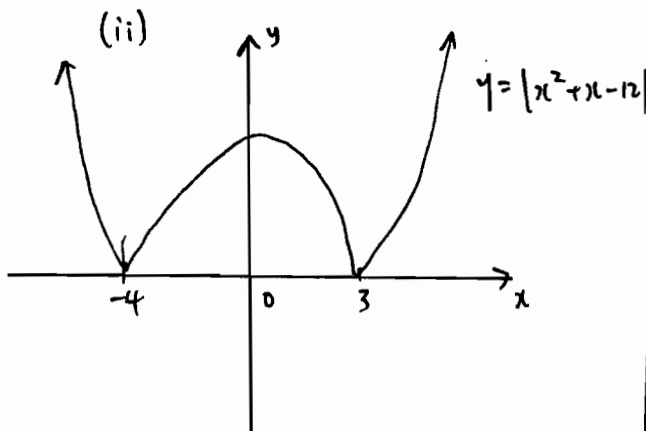
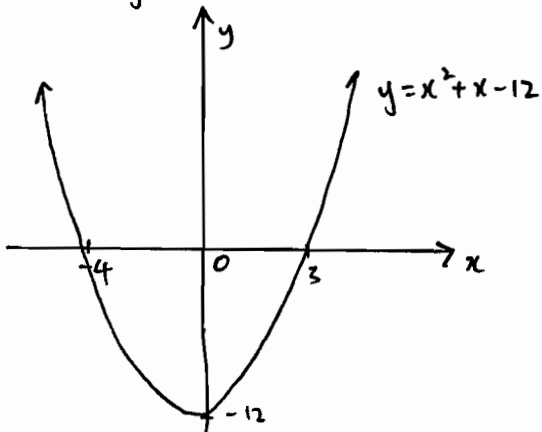
$$= \frac{2(2a + 2b - 5)}{a - b}$$

$$(c) i) \quad y = x^2 + x - 12 \Rightarrow y = (x + 4)(x - 3)$$

concave up

$$x \text{ int: } -4, 3$$

$$y \text{ int: } -12$$



$$(iii) \quad D: x \in \mathbb{R}$$

$$R: y \geq 0$$

Question 5

$$a) \quad \tan 300 = -\tan 60$$

$$= -\sqrt{3}$$

$$b) \quad 2\sin x + 1 = 0$$

$$\sin x = -\frac{1}{2}$$

$$\therefore 180^\circ \leq x \leq 360^\circ$$

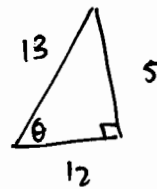
$$x = 210^\circ \text{ or } 330^\circ$$

$$c) \quad \angle BCA = 30^\circ$$

$$\therefore A = \frac{1}{2} \times 10 \times 6 \times \sin 30$$

$$= 15 \text{ cm}^2$$

d)



$$\sin \theta = -\frac{5}{13} \quad \tan \theta < 0$$

$\therefore \theta$ is in 4th quad

$\therefore \cos \theta$ is positive

$$\therefore \cos \theta = \frac{12}{13}$$

$$e) \quad \tan \theta + \cot \theta = \frac{1}{\sin \theta \cos \theta}$$

$$\text{LHS} = \tan \theta + \cot \theta$$

$$= \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \sin \theta}$$

$$= \frac{1}{\cos \theta \sin \theta}$$

$$= \text{RHS.}$$

Question 6

a) $y = 2x^4 - 8$
 $y' = 8x^3$

b) $y = (2x-1)^3$
 $y' = 3(2x-1)^2 \times 2$
 $= 6(2x-1)^2$

c) $y = x^2 \sqrt{x}$
 $= x^2 \cdot x^{\frac{1}{2}}$
 $= x^{\frac{5}{2}}$

$\therefore y' = \frac{5}{2} x^{\frac{3}{2}}$

d) $y = \frac{3}{2x^3}$
 $= \frac{3}{2} x^{-3}$
 $\therefore y' = -\frac{9}{2} x^{-4}$

e) $y = 5x^3(4x-3)^5$
 $u = 5x^3 \quad v = (4x-3)^5$
 $u' = 15x^2 \quad v' = 5(4x-3)^4 \times 4$
 $= 20(4x-3)^4$
 $\therefore y' = 5x^3 \times 20(4x-3)^4 + 15x^2(4x-3)^5$
 $= 100x^3(4x-3)^4 + 15x^2(4x-3)^5$
 $= 5x^2(4x-3)^4 [20x + 3(4x-3)]$
 $= 5x^2(4x-3)^4 [20x + 12x - 9]$
 $= 5x^2(4x-3)^4 [32x - 9]$

f) $y = \frac{x^2 - 2}{x+1}$

$u = x^2 - 2 \quad v = x+1$

$u' = 2x \quad v' = 1$

$y' = \frac{(x+1) \cdot 2x - (x^2 - 2)}{(x+1)^2}$

$= \frac{2x^2 + 2x - x^2 + 2}{(x+1)^2}$

$= \frac{x^2 + 2x + 2}{(x+1)^2}$

Question 7

(a) (i) In Δ 's ASB and ASD

AS is common

$\angle ASB = \angle ASD$ (right $\angle$'s, given)

$\angle BAS = \angle DAS$ (given)

$\therefore \Delta ASB \equiv \Delta ASD$ (AAS)

(ii) $DA = BA$ (corresponding sides of congruent Δ 's are equal)

(b) (i) $\angle SPQ = 104^\circ$

(ii) $\angle QRS = 104^\circ$ (opposite $\angle$'s of rhombus equal)

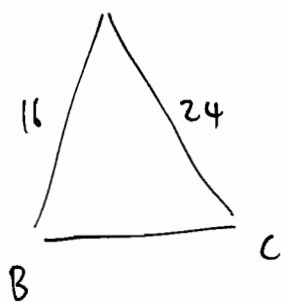
$\therefore \angle QRT = 76^\circ$ (straight angle)

(c) In Δ 's ADE and ABC

$\angle A$ is common

$\angle ADE = \angle ABC$ (corresponding $\angle$'s equal, $DE \parallel BC$)

$\therefore \Delta ADE \equiv \Delta ABC$ (equiangular)



$$\frac{16-x}{16} = \frac{18}{24}$$

$$16-x = \frac{18 \times 16}{24}$$

$$x = 16 - 12$$
$$= 4$$

8. a) $y = x^3 - 3x$

$$\frac{dy}{dx} = 3x^2 - 3$$

positive gradient when

$$\frac{dy}{dx} > 0$$

$$\therefore 3x^2 - 3 > 0$$

$$x^2 - 1 > 0$$

$$x > 1 \quad \text{or} \quad x < -1$$

b) $y = x^2 + 2x - 4$

$$\frac{dy}{dx} = 2x + 2$$

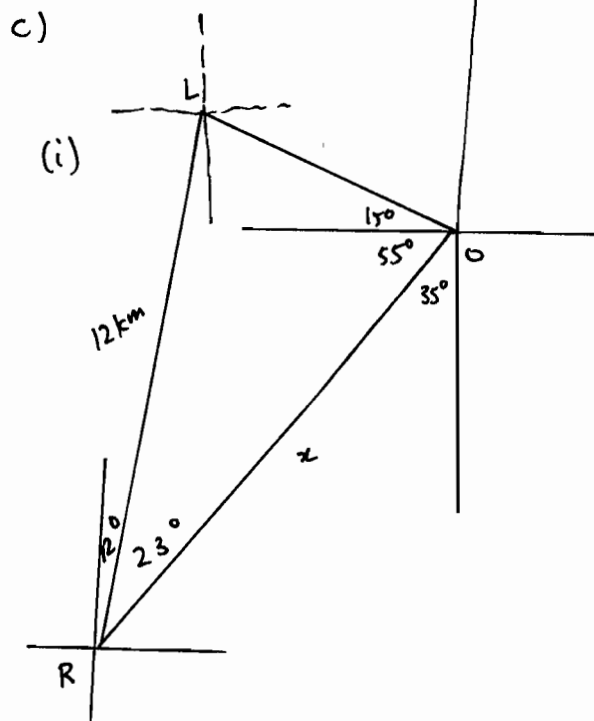
$$m_T = 2 \times 0 + 2 = 2$$

$$\therefore m_N = -\frac{1}{2}$$

$$\therefore y + 4 = -\frac{1}{2}(x - 0)$$

$$-2y - 8 = x$$

$$\therefore x + 2y + 8 = 0$$



$$(ii) \quad \angle RLO = 180 - 23 - 70^\circ$$
$$= 87^\circ$$

$$\frac{x}{\sin 87^\circ} = \frac{12}{\sin 70^\circ}$$

$$x = \frac{12 \sin 67^\circ}{\sin 70^\circ}$$

÷ 13 km

$$\begin{aligned} \text{(iii)} \quad & 180 + 12^\circ \\ & = 192^\circ \text{ T} \end{aligned}$$

Question 9

a) i) $6x^3 + 9x^2 - 3 = 3(x+1)^2(2x-1)$

$$\text{RHS} = 3(x+1)^2(2x-1)$$

$$= 3(x^2 + 2x + 1)(2x - 1)$$

$$= 3(2x^3 - x^2 + 4x^2 - 2x + 2x - 1)$$

$$= 3(2x^3 + 3x^2 - 1)$$

$$= 6x^3 + 9x^2 - 3$$

$$= \text{LHS}$$

(ii) α int: $-1, \frac{1}{2}$

b) $x=0$

$\therefore f(0) = -3$

$\therefore y \text{ int} = -3.$

c) $f(x) = 6x^3 + 9x^2 - 3$

$f'(x) = 18x^2 + 18x$

Stationary point(s) when $f'(x) = 0$

$18x^2 + 18x = 0$

$18x(x+1) = 0$

$\therefore x = 0 \text{ or } -1$

$\therefore y = -3 \text{ or } 0$

$\therefore (0, -3) \quad (-1, 0)$

Testing nature:

$f''(x) = 36x + 18$

$f''(0) = 18 > 0 \therefore \text{concave up}$

$\therefore \text{min at}$
 $(0, -3)$

$f''(-1) = -36 + 18$

$= -18 < 0 \therefore \text{concave down}$

$\therefore \text{max at}$
 $(-1, 0)$

$\therefore \text{minimum at } (0, -3)$

$\therefore \text{maximum at } (-1, 0)$

d) point of inflexion occurs

$f''(x) = 0$

$\Rightarrow 36x + 18 = 0$

$36x = -18$

$x = -\frac{1}{2}$

$y = -\frac{3}{2}$

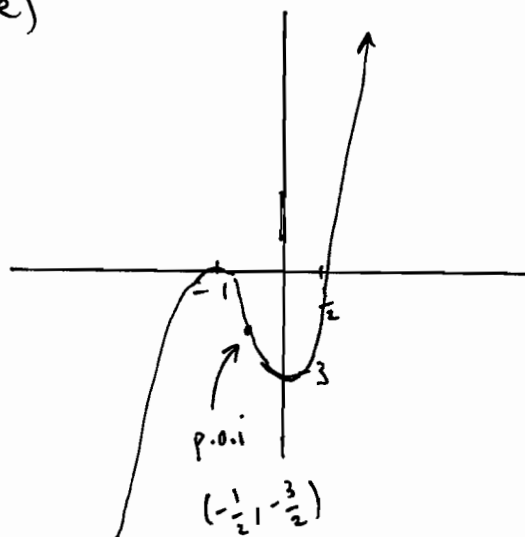
testing

x	-1	$-\frac{1}{2}$	0
$f''(x)$	-18	0	18
	$\wedge$	$\cdot$	$\vee$

$\therefore \text{change in concavity}$

$\therefore (-\frac{1}{2}, -\frac{3}{2}) \text{ is p.o.i.}$

e)



Question 10.

$$(i). SA = 2\pi r^2 + 2\pi rh$$

$$SA = 100\pi$$

$$\therefore 2\pi r^2 + 2\pi rh = 100\pi$$

$$r^2 + rh = 50$$

$$rh = 50 - r^2$$

$$h = \frac{50}{r} - r$$

$$ii) V = \pi r^2 h$$

$$= \pi r^2 \left(\frac{50}{r} - r \right)$$

$$= 50\pi r - \pi r^3$$

$$iii) \frac{dV}{dr} = 50\pi - 3\pi r^2$$

$$\frac{dV}{dr} = 0 \Rightarrow$$

$$50\pi - 3\pi r^2 = 0$$

$$3\pi r^2 = 50\pi$$

$$r^2 = \frac{50}{3}$$

$$r = \pm \sqrt{\frac{50}{3}}$$

$$r > 0$$

$$\therefore r = \sqrt{\frac{50}{3}}$$

$$= \frac{5\sqrt{2}}{\sqrt{3}}$$

testing

$$\frac{d^2V}{dr^2} = -6\pi r$$

$$= -6\pi \times \frac{5\sqrt{2}}{\sqrt{3}} < 0 \therefore \text{concave down}$$

$$\therefore \text{max occur at } r = \frac{5\sqrt{2}}{3}$$

$$\therefore V = 50\pi \times \left(\frac{5\sqrt{2}}{\sqrt{3}} \right) - \pi \left(\frac{5\sqrt{2}}{\sqrt{3}} \right)^3$$
$$\approx 428 \text{ cm}^3$$